

Analytical Mechanics

Selected Topics on Mathematical Physics | Autumn 2026

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I Prerequisites

Students are expected to be familiar with the following topics:

- Linear Algebra
- Complex Analysis
- Multivariable calculus

II Course Structure

The course meets twice per week for lectures. In addition, a weekly tutoring session is held to support work on the problem sheets; attendance is strongly encouraged, as the sheets form an integral part of the course.

	Format	Duration
Regular sessions	Theory and derivations	80 minutes
Tutoring session	Problem sheet workshop	90 minutes

Problem sheets should be attempted *before* the tutoring session. Hints and solutions are discussed in the tutoring session and collected shortly thereafter.

III Course Description

This course develops the fundamental principles of classical mechanics beyond the Newtonian framework, reformulating dynamics in terms of variational principles and generalised coordinates. The central aim is to reveal the deep mathematical structure underlying physical laws — structure that not only unifies and simplifies the solution of mechanical problems, but also provides the conceptual foundation for quantum mechanics, statistical mechanics, and field theory. The course opens with a quick review on Newtonian mechanics and then derives the Lagrangian and Hamiltonian formalisms using the powerful idea of d'Alembert's principle. We go through the relationship between symmetries and conservation laws giving a motivation for Noether's theorem (which we do not cover in this course, but we discuss a similar topic, namely, variational symmetries) — which we briefly discuss. The transition to the Hamiltonian formalism introduces phase space as the natural arena for mechanics, and canonical transformations are studied as the structure-preserving maps of that space.

IV Contents

1. **Newtonian Mechanics** — Newton's laws; free fall; drag forces; simple harmonic oscillator; central force; constraints; D'Alembert's principle.
2. **Lagrangian Mechanics** — Generalised coordinates; Lagrange's equations; Lagrangian of a system; holonomic constraints; symmetries and integrals of the motion; Jacobi's integral.
3. **Hamiltonian Mechanics** — Hamiltonian of a system; canonical equations; Hamilton's principle; least action principle; variational symmetries; Poisson brackets; canonical transformations.
4. **Rigid Body** — Configuration space $SO(3)$; Euler angles; the inertia tensor; Euler's equations; principal axis frame.

This syllabus is subject to change. Any modifications will be announced in class.

V References

The star rating reflects the mathematical sophistication of each text: (★) accessible with the course prerequisites; (★★) requires additional mathematical maturity; (★★★) intended for readers revisiting the subject — not recommended if this is your first encounter with the material.

- [1] Kasdin, N. J.; Paley, D. A. *Engineering Dynamics: A Comprehensive Introduction*. Princeton University Press, 2011. [**Primary** (★)]
- [2] Landau, L. D.; Lifshitz, E. M. *Mechanics*. Course of Theoretical Physics, Vol. 1. 3rd ed. Butterworth-Heinemann, 1976. [**Primary** (★)]
- [3] Torres del Castillo, G. F. *An Introduction to Hamiltonian Mechanics*. Birkhäuser, 2018. [**Primary** (★★)]
- [4] Goldstein, H.; Poole, C.; Safko, J. *Classical Mechanics*. 3rd ed. Addison-Wesley, 2002. [**Background reference** (★★★)]